NovaShell User Guide

Version 1.7.0

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1. Getting Started

Installation

Windows

1. Download `NovaShell_x.x.x_x64-setup.exe` from the Releases page
2. Run the installer and follow the wizard
3. Launch NovaShell from the Desktop shortcut or Start Menu

macOS

1. Download the `.dmg` file from Releases
2. Open the `.dmg` and drag NovaShell to Applications
3. If blocked: run `xattr -cr /Applications/NovaShell.app` in Terminal

Linux

1. Download `.AppImage`, `.deb`, or `.rpm` from Releases
2. For AppImage: `chmod +x NovaShell_*.AppImage && ./NovaShell_*.AppImage`
3. For deb: `sudo dpkg -i NovaShell_*_amd64.deb`

Interface Overview

When you first open NovaShell, you'll see:

```
+-------------------------------------------------+
|  [Title Bar]  NovaShell    [Theme dots] [_][x]    |
+-------------------------------------------------+
|  [Tab Bar]  Terminal 1  | Terminal 2  | [+]       |
+--------+----------------------------------------+
|        |                                         |
| Sidebar|           Terminal Area                  |
| [icons]|                                         |
|        |  user@machine:~$                        |
```

```
|        |                                          |
+--------+------------------------------------------+
|  [Status Bar]  Ready | PowerShell | main | 14:30  |
+-------------------------------------------------+
```

Key Areas:

- **Title Bar** — Theme selector (colored dots), window controls
- **Tab Bar** — Multiple terminal tabs, shell selector dropdown (+)
- **Sidebar** — 15 tool panels (click icons to switch)
- **Terminal** — Main terminal with full shell access
- **Status Bar** — Shell type, git branch, server location, clock

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2. Terminal Basics

Opening Tabs

- Click the **+** button in the tab bar to add a new terminal
- Click the dropdown arrow next to + to select a specific shell (PowerShell, CMD, Git Bash, WSL, etc.)

Split Panes

- Click the split button in the status bar (bottom right) to cycle through:
  - **No Split** — Single terminal
  - **Vertical Split** — Side by side
  - **Horizontal Split** — Top and bottom

Command Autocomplete

- Start typing a command (at least 2 characters)
- Suggestions appear automatically
- Press **Tab** to accept the highlighted suggestion
- Press **Up/Down** arrows to navigate suggestions

Search

- Press **Ctrl+F** to open the search overlay
- Type your search term and press Enter to find next
- Use the arrow buttons to navigate matches

Copy & Paste

- **Copy**: Select text with mouse, then Ctrl+C (or Ctrl+Shift+C)
- **Paste**: Ctrl+V (or Ctrl+Shift+V)
- **Right-click**: Copy if text selected, paste if not

Focus Mode

- Click the focus mode button in the title bar to hide all UI chrome
- Only the terminal remains visible
- Click again to restore the full interface

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3. SSH Connections

Adding a Connection

1. Click the **SSH** icon in the sidebar (monitor icon)
2. Click **Add Connection**
3. Fill in the details:
   - **Name** — Friendly name (e.g., "Web Server")
   - **Host** — IP address or hostname
   - **Port** — Default: 22
   - **Username** — SSH username
   - **Auth Method** — Password or Private Key
4. Click **Test** to verify the connection works
5. Click **Save**

Connecting

1. Click **Connect** on any saved connection
2. If password is needed and not saved:
   - Choose: **Save to Keychain** (permanent), **Remember for Session** (until app closes), or **Don't Save**
   - Enter your password
3. The SSH terminal opens with full xterm.js emulation

Password Storage Options

- **System Keychain** — Uses Windows Credential Manager, macOS Keychain, or Linux Secret Service. Persists across app restarts.
- **Session Memory** — Stored in RAM only. Gone when app closes. Never touches disk.
- **Don't Save** — Must enter password every time you connect.

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4. Cross-Server Navigation

This is one of NovaShell's most powerful features. Navigate between SSH servers as if they were directories.

Prerequisites

- Have at least one SSH connection configured in the SSH panel
- The connection must have credentials available (keychain, session password, or private key)

Basic Usage

Colon Syntax

```
# Jump to a server
cd webserver:/var/www

# Navigate to a specific directory on another server
cd dbserver:/home/admin

# Return to your local machine
cd local:~
```

Filesystem Syntax

```
# Same thing, different style
cd /servers/webserver/var/www
```

```
# List all available servers with their status
ls /servers
```

Output of `ls /servers`

```
/servers — NovaShell Virtual Server Filesystem

  local/              <- your machine
  webserver           connected  admin@192.168.1.10:22
  dbserver            disconnected  root@10.0.0.5:22
  staging             connected  deploy@staging.example.com:22

Usage:  cd /servers/webserver/var/www
        cd webserver:/var/www
        cd local:~
```

Multi-Hop Navigation

You can chain server jumps:

```
user@local:~$ cd webserver:/var/www
[NovaShell] Connecting to webserver (192.168.1.10)...
[NovaShell] Connected! Navigating to /var/www

user@webserver:/var/www$ cd dbserver:/home
[NovaShell] Connecting to dbserver (10.0.0.5)...
[NovaShell] Connected! Navigating to /home

user@dbserver:/home$ cd local:~
[NovaShell] Returning to local machine...
user@local:~$
```

StatusBar Breadcrumb

When navigating servers, the status bar shows your location:

```
[Server icon]  webserver  192.168.1.10  [local -> webserver]
```

Server Name Resolution

NovaShell matches server names by:

1. **Connection name** (case-insensitive) — e.g., "webserver", "WebServer"
2. **Host/IP** — e.g., "192.168.1.10"
3. **Connection ID** — Internal UUID

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5. SFTP File Transfer

Starting a Transfer

1. Click the **SFTP** icon in the sidebar
2. Select an SSH connection from the dropdown

3. Click **Connect**
4. Browse the dual-panel file browser:
    - **Left** — Local filesystem
    - **Right** — Remote filesystem

Transferring Files

- **Download**: Select files on the right (remote), click Download
- **Upload**: Select files on the left (local), click Upload
- Transfer progress appears in the transfer log at the bottom

File Management

On the remote server, you can:

- **Create folders** — Right-click > New Folder
- **Rename** — Right-click > Rename
- **Delete** — Right-click > Delete
- **Preview** — Click a text file to preview its contents (up to 1 MB)

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6. Infrastructure Monitor

Real-time monitoring dashboard for all your SSH servers.

Opening the Monitor

1. Click the **Infra Monitor** icon in the sidebar (gauge icon, last in the list)
2. You'll see 5 tabs: **Overview**, **Timeline**, **Alerts**, **Disk**, **Settings**

Starting Monitoring

1. Make sure servers are connected via the SSH panel
2. In the Overview tab, click the **Play** button on a server card
3. Metrics start updating every 10 seconds (configurable)

Understanding the Dashboard

Server Cards

Each connected server shows:

```
+-- webserver -------- Score: 87 ----------+
| CPU ########-- 78%  MEM ######-- 62%     |
| DSK #####----- 45%  NET up12MB dn4MB     |
| Load: 2.1 1.8 1.5                        |
| [CPU sparkline ~~~~~~~~~~]               |
| [MEM sparkline ~~~~~~~~~~]               |
| Top: nginx 12% | postgres 8%            |
+------------------------------------------+
```

- **Health Score** — 0-100 (green >70, yellow >40, red <40)
- **Metric Bars** — Color-coded by threshold (green/yellow/red)
- **Sparklines** — 60-point trend charts for CPU and Memory
- **Network Rate** — Calculated from consecutive snapshots (bytes/sec)

Compact Mode

Click the layout toggle button (top right) to switch to one-line-per-server view:

```
webserver   87  [||||] [||||] [|||]  [Stop]
dbserver    41  [||||] [||||] [|||]  [Stop]
```

Control Center Actions

Click a server card to expand the action panel:

When CPU is High

- **Show Processes** — Lists top 20 processes by CPU usage
- **Kill PID** — Enter a PID, then choose:
  - **SIGTERM** — Graceful shutdown (process can save state)
  - **SIGKILL** — Immediate termination (cannot be caught)
  - **Cancel** — Abort
- **Open Terminal** — Jump to the server via cross-server navigation

When Disk is High

- **Analyze Disk** — Opens the Disk Analyzer tab for deep analysis
- **Scan Large Files** — Finds files >100MB and largest directories
- **Scan Old Logs** — Shows rotated log files and their sizes

When Services Fail

- **Restart [service]** — Runs `systemctl restart`
- **Show All Failed** — Lists all failed systemd units

Alerts

The Alerts tab shows threshold violations:

- **Warning** (yellow) — Metric above warning threshold (default 80%)
- **Critical** (red) — Metric above critical threshold (default 95%)
- Click the checkmark to acknowledge an alert
- Alerts deduplicate: same metric/server won't re-alert within 60 seconds

Timeline

Chronological log of everything:

```
Today
  14:32  webserver  CPU at 92.3%
  14:33  webserver  Executed: top on webserver
  14:35  dbserver   Started monitoring dbserver
  14:38  webserver  Cleaned: System Logs
```

Settings

- **Polling Interval** — How often to collect metrics (5s, 10s, 15s, 30s, 60s)
- **Alert Thresholds** — Customize warning/critical levels for CPU, Memory, Disk

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7. Disk Analyzer

A CCleaner-style disk management tool built into the Infra Monitor.

Running a Scan

1. Go to **Infra Monitor > Disk** tab
2. Click a server name button to start scanning
3. Wait for the analysis to complete (usually 5-15 seconds)

Understanding the Results

Partitions

Each partition shows:

- **Donut chart** — Visual usage percentage
- **Progress bar** — With gradient coloring (green/yellow/red)
- **Details** — Used, Free, Total, device name, filesystem type

Largest Directories

```
/var/lib/docker  ################  12.4 GB  +2.3 GB
/var/log         #############     8.1 GB
/home            ########          6.2 GB
/usr             ######            4.8 GB
```

- Bars show relative size
- **Red delta** (+2.3 GB) appears when a directory grew significantly since last scan
- **Click any directory** to open a terminal at that path

Disk Growth Alerts

If any directory grew >500MB since the last scan, a red alert appears:

```
[!] Disk Growth Detected
/var/log     +2.3 GB
/tmp         +800 MB
vs scan at 14:32
```

Cleanup Categories

Each category can be expanded to show actions:

Category	What it scans	Safe to clean?
System Logs	Rotated files (.gz, .old, .1) older than 7 days	Yes
Systemd Journal	Journal logs older than 7 days	Yes
System Cache	/var/cache contents	Inspect first
Package Cache	apt/yum/dnf downloaded packages	Yes
Temporary Files	/tmp + /var/tmp files older than 7 days	Yes
Docker	Unused containers, networks, images	Yes (prune)
Core Dumps	Crash dumps from failed processes	Yes
Snap Packages	/var/lib/snapd/snaps	Manual only

Cleanup Workflow

The safe 3-step workflow:

Step 1: Inspect

Click **Inspect** on a category to see exactly which files will be affected:

```
Preview: System Logs
  2.1M  /var/log/syslog.3.gz
  1.8M  /var/log/auth.log.2.gz
  1.2M  /var/log/kern.log.1
  ...
```

Step 2: Review

Each category also has contextual actions:

- **View files** — List directory contents
- **Tail syslog** — See recent log entries
- **Show errors** — Recent journal errors
- **Docker images/volumes** — Docker-specific views

Step 3: Clean

Click **Clean**, then **Confirm Clean** to execute. The cleanup:

- Only deletes files matching safe criteria (rotated, >7 days old)
- Shows the command output so you see what was done
- Auto-rescans to show updated disk usage

Multi-Select Cleanup

1. Check the boxes next to categories you want to clean
2. A green summary bar appears at the bottom:

   ```
   3 selected -- ~8.3 GB
   Logs, Docker, Temp Files    [Clean Selected]
   ```

3. Click **Clean Selected** to batch-clean all checked categories

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8. Debug Console

Opening

Click the **Debug** icon in the sidebar (bug icon).

Features

- **Level Filters** — Toggle: Error, Warn, Info, Debug, Trace, Output
- **Source Filter** — Filter by terminal tab or SSH connection
- **Search** — Full-text search across all log entries
- **Pause/Resume** — Freeze the log view while still collecting
- **Export** — Save logs to a file
- **Persistence** — Logs saved to JSONL files with 7-day auto-cleanup

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9. AI Assistant

Prerequisites

- Install Ollama on your local machine
- Pull a model: `ollama pull llama3.2`

Using the AI

1. Click the **AI** icon in the sidebar (sparkles icon)
2. Select a model from the dropdown
3. Choose a mode:
    - **Chat** — General conversation
    - **Explain** — Explain code or errors
    - **Generate** — Generate shell commands
    - **Fix** — Fix error messages
4. Type your message and press Enter

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10. Hacking Mode

Activating

Click the **shield** icon in the title bar or sidebar. This:

- Switches to the matrix-green "hacking" theme
- Opens the Hacking panel
- Starts the security monitor

Tools Available

- **Recon** — Port scanning, banner grabbing, environment detection
- **Exploits** — Curated pentest script database
- **AI Copilot** — Security-focused AI analysis
- **Tools** — Hash calculator, Base64/URL/Hex encoder-decoder, reverse shell generator
- **History** — Encrypted session save/load
- **Alerts** — Real-time security event monitoring

Deactivating

Click the shield icon again to restore your previous theme.

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11. Keyboard Shortcuts

Shortcut	Action
Ctrl+F	Search in terminal
Ctrl+C	Copy (with selection) / SIGINT (without)
Ctrl+V	Paste from clipboard
Ctrl+Shift+C	Copy (alternative)
Ctrl+Shift+V	Paste (alternative)

Tab	Accept autocomplete suggestion
Right-click	Copy if selected, paste if not

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12. Troubleshooting

SSH Connection Fails

- Verify the server is reachable: `ping hostname`
- Check port 22 is open: `telnet hostname 22`
- Try **Test Connection** in the SSH panel before connecting
- Check authentication method (password vs key)

Terminal Looks Wrong

- Try switching themes (colored dots in title bar)
- If text is garbled, the shell may not support the terminal type — try a different shell

Infra Monitor Shows No Data

- Ensure the SSH connection is active (green status in SSH panel)
- The remote server must have standard Linux tools: `top`, `free`, `df`, `ps`
- Check the Timeline tab for error messages

Disk Analyzer Scan Hangs

- Scans use `timeout` to prevent hangs (10s for `du`, 5s for `find`)
- On very large filesystems, the scan may return partial data
- Use the **Rescan** button to retry

Auto-Update Not Working

- Check your internet connection
- Verify you have write permissions to the installation directory
- Download manually from the Releases page

NovaShell v1.7.0 - Built with Tauri v2 + React + TypeScript https://github.com/FomoDonkey/NovaShell